

Statement of Work
KERMIT Mounting Plate
Period of Performance: Task Order Issuance – June 30, 2027

Contract Overview

The International Space Station (ISS) Program has a requirement for a new mounting plate for the KEyence Research Microscope Testbed (KERMIT). The KERMIT is a commercial off-the-shelf microscope that provides researchers with fundamental microscope capabilities, including state-of-the-art imaging and analysis modules. KERMIT provides the ability to perform advanced microscopy during spaceflight and remotely from the ground. KERMIT is designed to streamline image collection and analysis through a single platform. The images collected from KERMIT are transferred to operators on the ground without having to wait for a return vehicle. KERMIT is currently installed into the Materials Science Research Rack (MSRR) on the ISS. The MSRR will soon be removed, and the KERMIT will be installed into a Basic EXPRESS Rack.

The scope shall encompass all activities required to design, fabricate, certify for flight, and install a new mounting plate for the KERMIT Head Unit and Control Box.

DOCUMENTS

Applicable Documents

Document #	Title
SSP 57000	Pressurized Payload Interface Requirements Document (IRD) for International Space Station (ISS)
SSP 57057	ISS Science, Technology, and Exploration Integration Flow
SSP 51721	ISS Safety Requirements Document
SSP 30599	ISS Safety Review Process
SSP 52054	CoFR Implementation Plan
OZ-23-048	PD Expectations Letter

Reference Documents

Document #	Title
KERMITMOUNT	KERMIT Mounting Plate Assembly
IC-4108	KERMIT Microscope Relocation Topology Assessment

REQUIREMENTS

The Contractor shall provide the necessary resources (except for those identified as Government Furnished Equipment/Property on this TO) to perform the following tasks/requirements to design, manufacture, certify for flight, and install the KERMIT Mounting Plate:

1. The Contractor shall be responsible for the design of the KERMIT Mounting Plate to fit in a Basic EXPRESS Rack (BER).
2. The KERMIT Mounting Plate shall be made to isolate the KERMIT Head Unit and Control Box from outside vibrations.
3. The Contractor shall be responsible for the design, development, and fabrication of the KERMIT Mounting Plate.
4. The Contractor shall produce fabrication and installation drawings of the KERMIT Mounting Plate.
5. The Contractor shall maintain a project schedule and report updates monthly at a minimum or as changes occur.
6. Console Reports – The Contractor shall submit a report of on-orbit operations within 48hrs of operations.
7. The Contractor shall define their proposed resources, including facility needs for ground testing and flight operations support in their proposal.
8. The Contractor shall provide engineering support, near real-time on-call support during KERMIT Mounting Plate installation, provide engineering analysis, and resolve anomalies and/or failures of KERMIT Mounting Plate. Anomaly resolution will require interactions with the Mission Evaluation Room (MER) and Payload Operations and Integration Center (POIC).
9. The Contractor shall define and deliver their proposed solution to addressing any on-orbit hazard controls for KERMIT Mounting Plate and plans to mitigate any hazards. This includes the completion of Safety Phase 0-3.
10. Contractor shall develop and deliver a Flight Certification that states all requirements (safety and interface) have been met and the payload/sample is safe and ready to launch and operate on the ISS for each flight.
 - a. Documentation that supports this Flight Certification shall be in the Contractor's best practices format and content and will be retained for the duration of the task order
 - b. The Flight Certification Statement shall meet the requirements defined in the OZ CoFR Plan for a payload/sample for launch and on-orbit operations.
11. The Contractor shall develop on-orbit operations products to support the installation and operation of the KERMIT Mounting Plate. The Contractor shall also be responsible for developing and executing procedures to mount the KERMIT Head Unit and KERMIT Control Box to the KERMIT Mounting Plate in the Basic EXPRESS rack (BER) on-orbit.

DELIVERABLES

No.	Item
1	Concept of Operations Initial due with TOP delivery Final due at ATP+30 days
2	Operations inputs delivered to the real-time operations process.
4	Contractor proposed safety data and associated products needed to close Hazard Reports delivered to ISS Hazard database.
5	Contractor proposed verification Products delivered to Veritas (or contractor's data repository if concurred by the NASA payload requirements verification process authority).
6	CoFR Statements delivered per SSP 52054 OZ CoFR Plan.
7	Flight Certification – kept at contractor facility for the duration of the task order. Delivered to government upon completion of the investigation.

DELIVERY INSTRUCTIONS

Hardware Deliveries

Authorized Government personnel shall accept all products delivered via DD Form 250 or 1149.

The Contractor shall deliver the flight hardware and documentation using the appropriate shipping and handling processes for flight hardware to the Cargo Mission Contractor at the Johnson Space Center for ISS Program launch processing and operations, F.O.B. destination to the following address, unless otherwise directed by NASA:

NASA JSC CARGO MISSION CONTRACT 80JSC017C0015
555 FORGE RIVER ROAD
SUITE 150
WEBSTER, TX 77598

GOVERNMENT FURNISHED PROPERTY (GFP) AND GOVERNMENT OWNED FACILITIES AND SERVICES

Government Services Requested

Service Requested:	Purpose of Service:	Location Requested:	Duration for Request:
EMI/EMC Testing	Verification Closeout	MSFC/Bldg 4708	1 week

Vibration Testing (optional)	Verification Closeout	MSFC	3 days
Off-gas Testing (optional)	Verification Closeout	White Sands	1 week
Labeling	Flight Labels	N/A	N/A
HOSC Services	Operations Support	MSFC/HOSC	2 days per use

Government Facilities Requested

Facility Requested:	Purpose of Facility:	Location of Facility:	Duration for Request:
EMI/EMC Testing	Verification Closeout	MSFC/Bldg 4708	1 week
Vibration Testing (optional)	Verification Closeout	MSFC	3 days
Off-gas Testing (optional)	Verification Closeout	White Sands	1 week
HOSC Services	Operations Support	MSFC/HOSC	2 days per use

Schedule and Milestone Payments

The Estimated Completion Date (ECD) of milestones and payments shall be as follows:

Milestone	Description	Estimated Date of Completion	Deliverable	Amount of Payment (\$)
1	Kickoff Meeting	OFI	Kickoff Charts, notes, and relevant information about the meeting	OFI
2	Critical Design Review (CDR)	OFI	Completion of a joint chaired CDR that includes NASA participation	OFI
3	Delivery and Certification of the KERMIT Slide Table	NLT 01/01/2027	Completion of the safety and verification for transport and on orbit installation	OFI
4	Final Report	NLT 6/30/2027	Final Report approved by NASA	OFI